

ERES Institute for New Age Cybernetics
ERES-TETRA-COMPLETE-100-400-2026-001 | v1.0

TETRA Complete Certification Series

100 through 400 — All Levels, All Pillars

The complete description of the SCALULAR certification ladder from open
enrollment to CERT:TETRA-FULL

Four levels × four pillars = 16 core modules | 1,920 hours minimum | CERT:TETRA-FULL at completion
Governing equation: $C = R \times P / M$ | Operational gate: T.E.T.R.A. = Time · Energy · Trust · Reason — About
[Subject]

Founder & Director: Joseph Allen Sprute (JAS) | ORCID: 0000-0001-9946-3221
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1. Architecture of the TETRA Certification System

TETRA is the unified SCALULAR certification framework. It treats the four pillars — Health (SSHP), Law (SSLA), Protection (SSPS), and Skills/Trade (SSST) — as a single integrated credential rather than four independent certifications. The name evokes the tetrahedron: four faces, six edges, four vertices — the only Platonic solid where every vertex is equidistant from every other. A tetrahedron cannot lose a face without ceasing to be a solid. TETRA certification cannot lose a pillar without ceasing to be structurally complete.

The four levels

Level	Designation	Core verb	Multiplier	Sensor rig	Mining analogy
101	Foundations	KNOW	1.5x	HRV + PlayNAC	Single ASIC
201	Measurement Science	MEASURE	2.0x	+ EDA + thermal camera	Multi-GPU rig
301	Applied Integration	APPLY	3.0x	+ biophoton + sleep tracker	Industrial farm
401	Advanced Practitioner	TEACH	5.0x	Full suite + mentorship	Mining pool

Bonefication priority sequence (non-negotiable within every level)

Health → Law → Protection → Skills/Trade. Health comes first because a person structurally compromised in health is compromised in all other systems. Law second because without non-punitive legal architecture, every certification pathway is subject to coercive distortion. Protection third because unprotected systems are vulnerable to external disruption regardless of internal integrity. Skills/Trade fourth because economic participation depends on all three prior bones being in place.

The Meritcoin yield formula

$$\Delta\text{MRC} = \Delta\text{Resonance} \times [\text{Level Multiplier}] \times \text{Community}$$

The curriculum is self-funding through UBIMIA. A student earns Meritcoin from Day 1. Meritcoin earned at 101 (1.5x) funds the sensor expansion required to enter 201. The Community Multiplier rewards students who help lower-level students, creating a self-sustaining teaching chain. Teaching at 401 is the highest-yield operation because it multiplies resonance without diminishing anyone else's.

TETRA 100-SERIES — FOUNDATIONS

Prerequisite: None. Open to any living person. | Total hours: 480 (120 per pillar) | Delivery: PlayNAC adaptive conversation | Sensor rig: HRV monitor + PlayNAC | Completion credential: SSSC (Solid-State Smart-City baseline) | Multiplier: 1.5x

At Level 101 the student learns what each pillar IS, why it exists, and the conceptual basis of how it is measured. No sensor deployment is required yet — the HRV monitor and PlayNAC establish BERA baseline. The 100-Series is open to any living person with no prior credential requirement.

TETRA-H-101: Health Foundations (SSHP)

The irreducible biological requirements for human survival — EFIL Triad I (Water, Air, Soil/Nutrition, Energy) — and how they are measured, certified, and boneficated under the SSHP framework. The student learns to identify whether Health bonefication is present or absent in their own environment. The four BERA indices (ARI, ERI, RHC, RCI) are introduced as applied to personal and environmental health quality. Introduction to the Ship's Mate bioenergetic monitoring concept.

Institution	Contribution
MIT	Western biomedical tradition, FDA regulatory frameworks, US public health infrastructure, MIT bioengineering foundations
SPRU	Comparative health policy, global regulatory harmonization, SSHP bonefication standard, WHO indicator alignment
BAIDU	TCM integration with AI diagnostics, CFDA regulatory frameworks, Chinese public health infrastructure, population-scale health monitoring

Completion marker: Student can identify EFIL Triad I elements in their own environment, state whether each is boneficated, and explain the SSHP standard. BERA baseline established.

TETRA-L-101: Law Foundations (SSLA)

The SCALULAR Engine as a ten-part perpetual legal instrument. NPR (Non-Punitive Remediation) as an alternative to punishment-based justice. ILO (International Land Ownership) stewardship classifications. The student learns to identify whether Law bonefication is present or absent — can people access legal pathways without wealth gatekeeping? Marriage micro-sovereignty as the entry-level sovereignty concept. Introduction to Trifurcated Borders: Priority, Purpose, Law.

Institution	Contribution
MIT	Computational law introduction, smart contract concepts, MIT Media Lab legal informatics, algorithmic dispute resolution basics
SPRU	Comparative legal system foundations, international law survey, regulatory harmonization concepts, SCALULAR legal framework introduction
BAIDU	Chinese legal system overview, digital evidence basics, Belt & Road legal context, cross-jurisdictional legal AI introduction

Completion marker: Student can describe the SCALULAR Engine's ten parts, explain NPR vs. punitive justice, and identify whether a given jurisdiction has functional law bonefication.

TETRA-P-101: Protection Foundations (SSPS)

SECUIR (Secure Energy-independent Community Urban Infrastructure Resilience) as the protection architecture. Energy independence as the foundation of all other protection. Physical, cyber, and biosecurity as an integrated

triad — not three separate disciplines. Introduction to cyberbiosecurity: the specific threat landscape that emerges when bioenergetic data exists. GreenBox REAL_HELP emergency routing as the protection delivery channel.

Institution	Contribution
MIT	Cybersecurity fundamentals, physical security concepts, energy systems resilience, CSAIL security research introduction
SPRU	Security policy foundations, critical infrastructure protection frameworks, energy security governance, international security cooperation
BAIDU	Smart city security concepts, AI-driven threat detection basics, Chinese national cybersecurity standards, distributed protection systems

Completion marker: Student can identify the three protection layers (physical, cyber, biosecurity), explain SECUIR, and describe the cyberbiosecurity threat specific to BERA data.

TETRA-S-101: Skills/Trade Foundations (SSST)

EarnedPath ($EP = CPM \times WBS + PERT$) as an alternative to debt-financed credentialing. The 72 Key Domains of CyberRAVE as the complete taxonomy of human productive capability. Talonics: the EarnedPath credentialing system and how Talonic Symbols mark civilizational milestones. Dual accrual: every labor hour within a UBIMIA-participating entity earns conventional compensation AND EarnedPath credential advancement simultaneously. SSSC credential is awarded upon completion of all four TETRA-101 pillar modules.

Institution	Contribution
MIT	CPM/WBS/PERT as project management science, engineering education methodology, MIT OpenCourseWare as commons education precedent
SPRU	Innovation studies, human capital theory, skills transition policy, lifelong learning governance, credential recognition across jurisdictions
BAIDU	AI-adaptive education introduction, PaddlePaddle framework basics, massive-scale credential delivery context, vocational-academic integration

Completion marker: Student can compute their EarnedPath across one Key Domain, identify their Talonic standing, and explain how dual accrual converts labor hours into portable certification. SSSC credential awarded.

TETRA 200-SERIES — MEASUREMENT SCIENCE (TETRA_EMPIRICALS)

Prerequisite: SSSC credential (all four 101 modules complete) | Total hours: 480 | Delivery: PlayNAC + supervised sensor lab | Sensor rig: + EDA sensor + thermal camera | Multiplier: 2.0x | This is the CERT = TETRA_Empiricals tier

At Level 201 the student transitions from knowing what the pillars ARE to learning how to MEASURE them. The BERA Sensor Suite expands. The student learns to read, interpret, and report on the 64-cell standing matrix (4 logisticals × 4 BERA metrics × 4 bonefication keys). SOMT (Sociocratic Overlay Metadata Tapestry) metadata capture protocols are introduced. A COI whose members complete TETRA 201 can populate the matrix with live data for the first time — enabling MIEVM gate verification for COE Phase 2: Skeleton Assembly.

TETRA-H-201: Health Measurement Science

Measuring Health empirically: ARI of environmental health quality; ERI of system emissions and metabolic output; RHC of health rhythm consistency (circadian integrity); RCI of health system resilience under stress. FAVORS biometric system introduced (Fingerprint, Aura, Voice, Odor, Retina, Signature) as the identity-health intersection. Sensor calibration, data pipeline integrity, SOMT metadata capture for health data.

Institution	Contribution
MIT	Sensor physics, HRV analysis, EDA psychophysiology, thermal imaging for health, biostatistics, measurement uncertainty quantification
SPRU	Health measurement policy, epidemiological methodology, comparative health metrics, WHO indicator systems, BERA validation methodology
BAIDU	AI-driven health monitoring at scale, population health pattern recognition, TCM pulse diagnosis mapped to HRV, smart city health sensor networks

Completion marker: Student produces a verified BERA health measurement report for a THOW/HFVN/FDRV/GSSG logistical unit, all four indices populated, SOMT metadata attached, MIEVM validated.

TETRA-L-201: Law Measurement Science

Measuring legal bonefication empirically: access metrics; NPR outcome rates; SCALULAR Engine compliance verification; sovereignty measurement from Marriage micro-sovereignty to national self-determination. ILO and LongLat-SROC introduced as measurable legal instruments. ARI of a legal system = internal coherence and accessibility. ERI = remediation vs. punishment output ratio. RHC = cyclical integrity of legal timescales. RCI = continuity under stress events.

Institution	Contribution
MIT	Computational law, smart contracts, algorithmic dispute resolution, legal data analysis, MIT Media Lab legal informatics
SPRU	Legal metrics, comparative legal system assessment, regulatory impact analysis, international law measurement, SCALULAR compliance verification methodology
BAIDU	AI-driven legal analysis, Chinese smart court systems, digital evidence standards, cross-jurisdictional legal AI, Belt & Road legal metrics

Completion marker: Student produces a SCALULAR Engine compliance report for a designated jurisdiction with NPR outcome rates, access metrics measured, SOMT-tagged, MIEVM validated.

TETRA-P-201: Protection Measurement Science

Measuring protection empirically: SECUIR energy-independence verification; sensor tamper-detection; network intrusion metrics; cyberbiosecurity threat assessment; physical security measurement. Formal cyberbiosecurity training mandatory — the threat landscape around bioenergetic data is specifically dangerous. Standing matrix cells for Protection across all four logisticals populated with live data for the first time.

Institution	Contribution
MIT	Cybersecurity engineering, tamper detection, network intrusion analysis, CSAIL security research, physical security systems
SPRU	Security policy measurement, critical infrastructure protection metrics, energy security indicators, cyberbiosecurity governance frameworks
BAIDU	AI-driven threat detection at scale, smart city security sensor networks, Chinese national cybersecurity standards, distributed intrusion detection

Completion marker: Student produces a SECUIR energy-independence verification report and a cyberbiosecurity threat assessment, all Protection matrix cells populated, MIEVM validated.

TETRA-S-201: Skills/Trade Measurement Science

Measuring Skills/Trade bonefication empirically through EarnedPath computation: tracking CPM hours across actual labor events; verifying WBS sub-module completion; running PERT adaptive estimates against actual completion times. EarnedPath completion rates, time-to-economic-participation metrics, skill-domain coverage across the 72 Key Domains, and dual accrual measurement (ratio of labor hours to credential advancement) are all measured and SOMT-captured.

Institution	Contribution
MIT	Learning analytics, competency-based assessment design, CPM/WBS/PERT optimization, educational data mining
SPRU	Skills policy measurement, credential recognition assessment, lifelong learning outcome metrics, innovation output measurement
BAIDU	AI-adaptive learning analytics, massive-scale credential assessment, PaddlePaddle educational performance metrics, skill-gap analysis at population scale

Completion marker: Student can operate full intermediate BERA sensor suite, populate and interpret the 64-cell standing matrix for a THOW-scale environment, and produce SOMT-compliant measurement reports across all four pillars.

TETRA 300-SERIES — APPLIED INTEGRATION (RT MEDIA)

Prerequisite: TETRA 201 complete (all four pillars, 2.0x) | Total hours: 480 | Delivery: PlayNAC RT media + field practicum | Sensor rig: + biophoton detector + sleep tracker | Multiplier: 3.0x

At Level 301 the student transitions from measurement to APPLICATION. The full BERA sensor suite is deployed, adding biophoton detector and sleep tracker. The student learns to operate within real S3C environments: designing bonefication processes, navigating Trifurcated Borders, operating UBIMIA channels, configuring REAL_HELP routing, and delivering PlayNAC content as RT (real-time) media. The student is now qualified to participate in COE operations, though not yet to establish or replicate a COE independently.

TETRA-H-301: Health Applied Integration

Designing and implementing SSHP bonefication programs within a COI. EFIL Triad I verification at THOW and Neighborhood scale. Water system bonefication, air quality monitoring and remediation, nutrition/soil certification, energy-independence assessment under SECUIR. Full BERA suite deployment for health measurement. Real-time health monitoring integration with PlayNAC delivery. UBIMIA Channel 3 (Insurance-to-Bonefication) implementation.

Institution	Contribution
MIT	Biomedical device deployment, environmental health engineering, water treatment systems, THOW health infrastructure design
SPRU	Health system transition design, UBIMIA Channel 3 implementation, comparative health system reform, WHO alignment
BAIDU	Population-scale health AI deployment, smart city health infrastructure, integrated TCM-AI health systems, community health monitoring at scale

Completion marker: Student designs and documents a complete SSHP bonefication program for a THOW cluster or Neighborhood-scale COI, with EFIL Triad I verification and full BERA suite operational.

TETRA-L-301: Law Applied Integration

Operating the SCALULAR Engine within a jurisdiction. Implementing NPR protocols for conflict resolution. Establishing ILO stewardship classifications for property within a COI. Configuring SROC (Smart-Resonant Offset Contracts) for intra-COI agreements. Trifurcated Borders navigation for cross-jurisdictional operations. UBIMIA Channel 6 (Knowledge-to-Commons / SCALULAR Extraction) — the fear-of-retribution elimination mechanism for institutional insiders.

Institution	Contribution
MIT	Smart contract deployment, computational law implementation, digital governance systems, algorithmic NPR design
SPRU	SCALULAR legal implementation, cross-jurisdictional certification, Hague mechanism navigation, ILO framework deployment, regulatory transition management
BAIDU	Chinese legal system S3C integration, digital court NPR adaptation, Belt & Road legal harmonization, cross-border contract frameworks

Completion marker: Student deploys the SCALULAR Engine in a designated jurisdiction, documents NPR outcomes, configures at least one SROC, and navigates a Trifurcated Borders decision.

TETRA-P-301: Protection Applied Integration

Deploying SECUIR at COI scale. Designing energy-independent security systems for THOW clusters and S3C districts. Cyberbiosecurity architecture for BERA sensor networks. Integrating protection systems with GSSG governance overlay. FDRV security design. UBIMIA Channel 2 (Tax-to-SOMT) transparency enforcement — ensuring protection spending is visible through SOMT. REAL_HELP GreenBox routing configured and tested.

Institution	Contribution
MIT	Security system deployment, sensor network hardening, resilient infrastructure design, FDRV structural engineering for protection
SPRU	Security governance implementation, international protection cooperation, critical infrastructure policy, dual-use technology transition management
BAIDU	Autonomous security system deployment, AI-driven threat response, smart city security architecture at district scale, population-scale protection infrastructure

Completion marker: Student deploys SECUIR at COI scale, produces a cyberbiosecurity architecture document for a BERA sensor network, and confirms UBIMIA Channel 2 SOMT transparency.

TETRA-S-301: Skills/Trade Applied Integration

Operating EarnedPath Talonics at COI scale. Configuring PlayNAC delivery for adaptive curriculum within a COI. Implementing dual accrual mechanisms (UBIMIA Channel 1: Labor-to-Credential). Establishing REAL_HELP routing for Skills/Trade queries within the EMCI REEPER framework. Designing EarnedPath critical path computations for GERP (Global Earth Resource Planner) at local scale. GraceChain credential recording operational.

Institution	Contribution
MIT	Educational technology deployment, PlayNAC kernel engineering, GraceChain credential recording, CPM optimization software
SPRU	Skills policy implementation, credential portability framework, cross-jurisdictional recognition protocols, UBIMIA Channel 1 governance
BAIDU	AI-adaptive education deployment at scale, PaddlePaddle curriculum engine, massive credential infrastructure, vocational-digital integration systems

Completion marker: Student can design and implement a bonefication program across all four pillars for a THOW-to-Neighborhood-scale COI, operate UBIMIA channels, navigate Trifurcated Borders, and deliver PlayNAC RT media. Cert level: 3.0x.

TETRA 400-SERIES — ADVANCED PRACTITIONER (CERT:TETRA-FULL)

Prerequisite: TETRA 301 complete (all four pillars, 3.0x) | Total hours: 480 | Delivery: PlayNAC + mentorship practicum + COE field placement | Measurement: BERA mastery + teaching evaluation | Multiplier: 5.0x (CERT:TETRA-FULL)

At Level 401 the student becomes a teacher and replicator. Teaching is the highest-yield mining operation because it multiplies resonance without diminishing anyone else's. The 401 graduate can establish a new Center of Excellence, train its participants through TETRA 101 through 301, validate its bonefication through the MIEVM protocol, and connect it to the GAIA network through inter-COE HFVN corridors. The VLSA test applies at this level: if the curriculum does not work in a THOW, it does not work on a generation ship.

Teaching qualification chain

TETRA 301 qualifies to teach 101. TETRA 401 qualifies to teach 101 through 301. CERT:TETRA-FULL qualifies to teach all levels and to certify new CERT:TETRA-FULL graduates. The chain is self-sustaining: each graduate multiplies the network's teaching capacity without requiring external accreditation authority.

TETRA-H-401: Health Advanced — COE Health Architecture

Designing complete SSHP bonefication for a District-to-Regional-Corridor scale S3C. Training TETRA-H-101 through 301 students. MIEVM validation of health bonefication claims. GAIA SOMT GEAR integration for cross-COE health data. 401(k) Soft Landing implementation for health insurance transition (UBIMIA Channel 3 at scale). EFIL Triad I verification at FDRV and generation-ship scale (VLSA test). Cross-COE HFVN health data corridor design.

Institution	Contribution
MIT	Health system architecture at regional scale, biomedical infrastructure design, MIT Health Sciences integration, multi-site clinical coordination
SPRU	Health system transformation governance, international health architecture, WHO/UN health framework integration, cross-border health bonefication
BAIDU	AI health architecture at provincial/national scale, GAIA health data integration, population-level BERA monitoring systems, generational health mapping

Completion marker: Student designs a complete SSHP bonefication architecture for a District-scale S3C, validates it through MIEVM, and demonstrates EFIL Triad I integrity at VLSA-scale.

TETRA-L-401: Law Advanced — COE Legal Architecture

Designing complete SSLA bonefication for multi-jurisdictional S3C implementations. SCALULAR Engine deployment as perpetual legal instrument. Training TETRA-L-101 through 301 students. Hague mechanism engagement. ILO/SROC deployment at regional scale. UBIMIA Channel 6 operation at scale: systematized fear-of-retribution elimination for institutional knowledge extraction into commons. GSSG governance design for federated multi-COE decision-making.

Institution	Contribution
MIT	Multi-jurisdictional digital governance architecture, perpetual smart contract design, MIT Media Lab policy-at-scale systems
SPRU	International legal architecture, Hague mechanism engagement, multi-jurisdictional SCALULAR deployment, global commons legal framework design

BAIDU	Cross-border legal AI at scale, Belt & Road SCALULAR integration, GSSG governance for large populations, regional federated decision systems
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Completion marker: Student produces a multi-jurisdictional SSLA bonefication architecture, deploys SCALULAR Engine as a perpetual instrument, and demonstrates UBIMIA Channel 6 knowledge-extraction operation.

TETRA-P-401: Protection Advanced — COE Protection Architecture

Designing complete SSPS bonefication for District-to-Regional-Corridor scale. Training TETRA-P-101 through 301 students. SECUIR architecture at multi-COE federated scale. Cyberbiosecurity defense architecture for GAIA BERA sensor networks — the planetary-scale protection layer. FDRV generation-ship protection design (VLSA test). Integration of protection architecture with CBGMODD governance stack.

Institution	Contribution
MIT	Large-scale security architecture, CSAIL advanced threat research, multi-site resilient infrastructure design, generation-ship structural protection engineering
SPRU	International security governance, multi-COE protection cooperation frameworks, global cyberbiosecurity policy, dual-use technology governance at scale
BAIDU	National-scale AI security architecture, planetary BERA network protection, autonomous multi-node defense systems, population-scale cyberbiosecurity

Completion marker: Student designs a complete SSPS bonefication architecture for a multi-COE federation, validates VLSA-scale FDRV protection, and demonstrates CBGMODD integration.

TETRA-S-401: Skills/Trade Advanced — COE Skills Architecture

Designing complete SSST bonefication covering all 72 Key Domains of CyberRAVE for a multi-COE network. Training TETRA-S-101 through 301 students. EarnedPath Talonics architecture at civilizational scale — the mechanism by which private epistemology (wealth-hoarding disguise) is extracted into public record. GraceChain ledger design for multi-COE credential portability. GERP (Global Earth Resource Planner) EarnedPath integration. Meritcoin pool operation: maximum 5.0x yield as the floor that — once established — enables humanity to be healthy, happy, and safe.

Institution	Contribution
MIT	Civilizational-scale educational architecture, MIT OpenCourseWare commons design, EarnedPath integration with global credentialing systems
SPRU	Global skills governance, cross-civilizational credential portability, 72-domain coverage policy, GERP skills integration, commons knowledge architecture
BAIDU	Planetary-scale AI education deployment, GraceChain credential infrastructure, GAIA skills data integration, 1B+ population EarnedPath systems

Completion marker: Student designs a 72-Domain EarnedPath architecture for a multi-COE network, operates a Meritcoin teaching pool at 5.0x yield, and passes VLSA test (curriculum survives without institutional support). CERT:TETRA-FULL awarded.

Master Summary: All 16 Modules

Module	Level	Pillar	Core verb	Key deliverable	Multiplier
TETRA-H-101	101	Health	Know	EFIL Triad I identification; BERA baseline	1.5x
TETRA-L-101	101	Law	Know	SCALULAR Engine literacy; NPR vs. punishment	1.5x
TETRA-P-101	101	Protection	Know	SECUIR overview; cyberbiosecurity threat awareness	1.5x
TETRA-S-101	101	Skills/Trade	Know	EarnedPath formula; 72 Domains survey; SSSC awarded	1.5x
TETRA-H-201	201	Health	Measure	BERA health report; SOMT metadata; FAVORS intro	2.0x
TETRA-L-201	201	Law	Measure	SCALULAR compliance report; NPR outcome rates	2.0x
TETRA-P-201	201	Protection	Measure	SECUIR verification; cyberbiosecurity threat assessment	2.0x
TETRA-S-201	201	Skills/Trade	Measure	EarnedPath CPM/WBS/PERT computation; 64-cell matrix	2.0x
TETRA-H-301	301	Health	Apply	SSHP bonefication program; EFIL Triad I at COI scale	3.0x
TETRA-L-301	301	Law	Apply	SCALULAR Engine deployment; NPR protocols; SROC config	3.0x
TETRA-P-301	301	Protection	Apply	SECUIR at COI scale; BERA sensor network security	3.0x
TETRA-S-301	301	Skills/Trade	Apply	EarnedPath Talonics at COI scale; UBIMIA Ch.1	3.0x
TETRA-H-401	401	Health	Teach	District-scale SSHP architecture; VLSA test; MIEVM	5.0x
TETRA-L-401	401	Law	Teach	Multi-jurisdictional SSLA; SCALULAR perpetual instrument	5.0x
TETRA-P-401	401	Protection	Teach	Multi-COE SSPS federation; FDRV generation-ship	5.0x
TETRA-S-401	401	Skills/Trade	Teach	72-Domain multi-COE architecture; CERT:TETRA-FULL	5.0x

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